DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY
UTTAR PRADESH, LUCKNOW



Evaluation Scheme & Syllabus

For

B.Tech. 4th Year

Artificial Intelligence & Data Science

(Effective from the Session: 2024-25)

B. TECH 4th Year

ARTIFICIAL INTELLIGENCE & DATA SCIENCE

SEMESTER- VII													
Sl. No.	**Subject Codes**	**Subject**	**Periods**			**Evaluation Scheme**				**End Semester**		**Total**	**Credit**
			L	**T**	**P**	**CT**	**TA**	**Total**	**PS**	**TE**	**PE**		
1	KHU701/KHU702	HSMC -1 / HSMC-2	3	0	0	30	20	50		100		150	3
2	Departmental Elective-IV	Departmental Elective-IV	3	0	0	30	20	50		100		150	3
3	Departmental Elective-V	Departmental Elective-V	3	0	0	30	20	50		100		150	3
4	KOE07X	Open Elective-II	3	0	0	30	20	50		100		150	3
5	KCS751A	Departmental Elective Lab**	0	0	2				25		25	50	1
6	KCS752	Mini Project or Internship Assessment*	0	0	2				50			50	1
7	KCS753	Project	0	0	8				150			150	4
8		MOOCs (Essential for Hons. Degree)											
		Total	**12**	**0**	**12**							**850**	**18**

***The Mini Project or internship (4 - 6 weeks) conducted during summer break after VI semester and will be assessed during VII semester.

**Department may conduct one Lab of based on the elective subject chosen by the students in either Departmental Elective-IV or Departmental Elective-V

SEMESTER- VIII													
Sl. No.	**Subject Codes**	**Subject**	**Periods**			**Evaluation Scheme**				**End Semester**		**Total**	**Credit**
			L	**T**	**P**	**CT**	**TA**	**Total**	**PS**	**TE**	**PE**		
1	KHU801/KHU802	HSMC-2#/HSMC-1#	3	0	0	30	20	50		100		150	3
2	KOE08X	Open Elective-III	3	0	0	30	20	50		100		150	3
3	KOE09X	Open Elective-IV	3	0	0	30	20	50		100		150	3
4	KCS 851	Project	0	0	18				100		300	400	9
5		MOOCs (Essential for Hons. Degree)											
		Total	**9**	**0**	**18**							**850**	**18**

Departmental Elective-IV

1. KDS071 Artificial Intelligence
2. KCS072 Natural language processing
3. KDS073 Text Analytics
4. KAD071 Prompt Engineering and AI Application Design
5. KAI075 Data Warehousing and Data Mining
6. KAI076 Time series analysis and Forecasting
7. KAI078 Nature-Inspired Computing

Departmental Elective-V

1. KDS078 Deep Learning
2. KDS079 Service Oriented Architecture
3. KAD072 Large Language Models (LLMs) and Their Applications
4. KAD073 Robotics and Automation
5. KAD074 Cognitive computing
6. KCS710 Quantum Computing
7. KAD075 Machine Learning & Network Security

B. TECH 4th Year

ARTIFICIAL INTELLIGENCE & DATA SCIENCE

KDS071	Artificial Intelligence	
Course Outcome (CO)		**Bloom's Knowledge Level (KL)**
At the end of course, the student will be able to understand		
CO 1	Understand the basics of the convex optimization.	K_2
CO 2	Understand the different Gradient-based methods.	K_2, K_3
CO 3	Can implement Newton's method and L-BFGS solvers for convex optimization problems,	K_3 , K_4
CO 4	Can identify the trade-offs inherent in using first-order vs. second-order solvers for optimization problems arising in machine learning.	K_2 , K_3
CO 5	Demonstrate competence with probability theory/statistics needed to formulate and solve machine learning problems.	K_2 , K_4

DETAILED SYLLABUS		**3-0-0**
Unit	**Topic**	**Proposed Lecture**
I	**Basics of convex optimization** Convex sets, convexity-preserving operations, examples of convex programs (linear programming (LP), second-order cone programming (SOCP), semidefinite programming (SDP)), convex relaxation, KKT conditions, duality	**09**
II	**Gradient-based methods** Gradient descent, subgradient, mirror descent, Frank–Wolfe method, Nesterov's accelerated gradient method, ODE interpretations, dual methods, Nesterov's smoothing, proximal gradient methods, Moreau–Yosida regularization	**09**
III	**Operator splitting methods** Augmented Lagrangian methods, alternating direction method of multipliers (ADMM), monotone operators, Douglas–Rachford splitting, primal and dual decomposition	**09**
IV	**Stochastic and nonconvex optimization** Dual averaging, Polyak–Juditsky averaging, stochastic variance reduced gradient (SVRG), Langevin dynamics, escaping saddle points, landscape of nonconvex problems, deep learning	**09**
V	**Two Use Case of ML optimization Techniques**	**04**

Text books:

1. Stephen Boyd and Lieven Vandenberghe's book: Convex Optimization
2. Nesterov's old book: Introductory Lectures on Convex Optimization: A Basic Course
3. Neal Parikh and Stephen Boyd's monograph: Proximal Algorithms
5. S´ebastien Bubeck's monograph: Convex Optimization: Algorithms and Complexity
6. Moritz Hardt's Berkeley EE 227C course note
7. Prateek Jain and Purushottam Kar's survey on nonconvex optimization
8. Linear Algebra and Learning from Data, Gilbert Strang
9. Convex Optimisation by Stephen Boyd
10. Optimisation for Machine Learning by Suvrit Sra, MIT Press.

KCS072	Natural Language Processing	
Course Outcome (CO)		**Bloom's Knowledge Level (KL)**
At the end of course , the student will be able :		
CO 1	To learn the fundamentals of natural language processing	K_1 , K_2
CO 2	To understand the use of CFG and PCFG in NLP	K_1 , K_2
CO 3	To understand the role of semantics of sentences and pragmatic	K_2
CO 4	To Introduce Speech Production And Related Parameters Of Speech.	K_1 , K_2
CO 5	To Show The Computation And Use Of Techniques Such As Short Time Fourier Transform, Linear Predictive Coefficients And Other Coefficients In The Analysis Of Speech.	K_3, K_4

DETAILED SYLLABUS		**3-0-0**
Unit	**Topic**	**Proposed Lecture**
I	**INTRODUCTION:** Origins and challenges of NLP – Language Modeling: Grammar-based LM, Statistical LM – Regular Expressions, Finite-State Automata – English Morphology, Transducers for lexicon and rules, Tokenization, Detecting and Correcting Spelling Errors, Minimum Edit Distance **WORD LEVEL ANALYSIS :** Unsmoothed N-grams, Evaluating N-grams, Smoothing, Interpolation and Backoff – Word Classes, Part-of-Speech Tagging, Rule-based, Stochastic and Transformation-based tagging, Issues in PoS tagging – Hidden Markov and Maximum Entropy models.	**08**
II	**SYNTACTIC ANALYSIS:** Context Free Grammars, Grammar rules for English, Treebanks, Normal Forms for grammar – Dependency Grammar – Syntactic Parsing, Ambiguity, Dynamic Programming parsing – Shallow parsing – Probabilistic CFG, Probabilistic CYK, Probabilistic Lexicalized CFGs – Feature structures, Unification of feature structures.	**08**
III	**SEMANTICS AND PRAGMATICS**: Requirements for representation, First-Order Logic, Description Logics – Syntax-Driven Semantic analysis, Semantic attachments – Word Senses, Relations between Senses, Thematic Roles, selectional restrictions – Word Sense Disambiguation, WSD using Supervised, Dictionary & Thesaurus, Bootstrapping methods – Word Similarity using Thesaurus and Distributional methods.	**08**
IV	**BASIC CONCEPTS of Speech Processing :** Speech Fundamentals: Articulatory Phonetics – Production And Classification Of Speech Sounds; Acoustic Phonetics – Acoustics Of Speech Production; Review Of Digital Signal Processing Concepts; Short-Time Fourier Transform, Filter-Bank And LPC Methods.	**08**
V	**SPEECH-ANALYSIS**: Features, Feature Extraction And Pattern Comparison Techniques: Speech Distortion Measures– Mathematical And Perceptual – Log–Spectral Distance, Cepstral Distances, Weighted Cepstral Distances And Filtering, Likelihood Distortions, Spectral Distortion Using A Warped Frequency Scale, LPC, PLP And MFCC Coefficients, Time Alignment And Normalization – Dynamic Time Warping, Multiple Time – Alignment Paths. **SPEECH MODELING** : Hidden Markov Models: Markov Processes, HMMs – Evaluation, Optimal State Sequence – Viterbi Search, Baum-Welch Parameter Re-Estimation, Implementation Issues.	**08**

Text books:

1. Daniel Jurafsky, James H. Martin—Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics and Speech, Pearson Publication, 2014.
2. Steven Bird, Ewan Klein and Edward Loper, —Natural Language Processing with Python, First Edition, OReilly Media, 2009.
3. Lawrence Rabiner And Biing-Hwang Juang, "Fundamentals Of Speech Recognition", Pearson Education, 2003.
4. Daniel Jurafsky And James H Martin, "Speech And Language Processing – An Introduction To Natural Language Processing, Computational Linguistics, And Speech Recognition", Pearson Education, 2002.
5. Frederick Jelinek, "Statistical Methods Of Speech Recognition", MIT Press, 1997.
6. Breck Baldwin, —Language Processing with Java and LingPipe Cookbook, Atlantic Publisher, 2015.
7. Richard M Reese, —Natural Language Processing with Java, OReilly Media, 2015.
8. Nitin Indurkhya and Fred J. Damerau, —Handbook of Natural Language Processing, Second Edition, Chapman and Hall/CRC Press, 2010.
9. Tanveer Siddiqui, U.S. Tiwary, —Natural Language Processing and Information Retrieval, Oxford University Press, 2008.

KDS073	Text Analytics	
Course Outcome (CO)		**Bloom's Knowledge Level (KL)**
At the end of course , the student will be able to understand		
CO 1	Engage in continuous reflective learning in the context of technology and scientific advancement.	K2
CO 2	To learn understand the use of text analytics and Processing	K2, K3
CO 3	To understand the role of Text mining modeling using NLTK	K3 , K4
CO 4	To Introduce Speech Production And Related Parameters Of Speech.	K2 , K3
CO 5	To Show The Computation And Use Of Techniques Such As Short Time Fourier Transform, Linear Predictive Coefficients And Other Coefficients In The Analysis Of Speech.	K2 , K4

DETAILED SYLLABUS		**3-0-0**
Unit	**Topic**	**Proposed Lecture**
I	Introductory overview of Text Mining , Introductory Thoughts , Data Mining vs. Text Mining ,Text Mining and Text Characteristics , Predictive Text Analytics ,Text Mining Problems ,Prediction & Evaluation	**08**
II	Text mining modeling using NLTK , Text Corpus , Sentence Tokenization , Word Tokenization , Removing special Characters , Expanding contractions, Removing Stopwords , Correcting words: repeated characters , Stemming & lemmatization ,Part of Speech Tagging.	**08**
III	Feature Extraction, Bag of words model, TF-IDF model, Text classification problem, Building a classifier using support vector machine.	**08**
IV	Introduction to natural language processing (NLP) Linguistics Essentials. Foundations of text processing: tokenization, stemming, stopwords, lemmatization, part-of-speech tagging, syntactic parsing.	**08**
V	Word Level Analysis : Unsmoothed N-grams, Evaluating N-grams, Smoothing, Interpolation and Backoff – Word Classes, Part-of-Speech Tagging, Rule-based, Stochastic and Transformation-based tagging, Issues in PoS tagging – Hidden Markov and Maximum Entropy models.	**08**

Text books:

1. Fundamentals of Predictive Text Mining by Sholom M. Weiss, Nitin Indurkhya, & Tong Zhang (2010/2015)
2. Text Analytics with Python: A Practical Real-World Approach to Gaining Actionable Insights from Your Data by Dipanjan Sarkar (2016)
3. Daniel Jurafsky, James H. Martin―Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics and Speech, Pearson Publication, 2014.
4. Steven Bird, Ewan Klein and Edward Loper, ―Natural Language Processing with Python, First Edition, OReilly Media, 2009.
5. Lawrence Rabiner And Biing-Hwang Juang, "Fundamentals Of Speech Recognition", Pearson Education, 2003.
6. Daniel Jurafsky And James H Martin, "Speech And Language Processing – An Introduction To Natural Language Processing, Computational Linguistics, And Speech Recognition", Pearson Education, 2002.
7. Frederick Jelinek, "Statistical Methods Of Speech Recognition", MIT Press, 1997.
8. Breck Baldwin, ―Language Processing with Java and LingPipe Cookbook, Atlantic Publisher, 2015.
9. Richard M Reese, ―Natural Language Processing with Java, OReilly Media, 2015.

KAD071	Prompt Engineering and AI Application Design	
Course Outcome (CO)		**Bloom's Knowledge Level (KL)**
At the end of course , the student will be able to understand		
CO 1	Understand the fundamentals of prompt engineering and its importance in AI applications.	K2 , K3
CO 2	Develop effective prompts for various AI models, including GPT and other LLMs.	K1 , K2
CO 3	Apply prompt engineering techniques to optimize model outputs for different use cases.	K4
CO 4	Analyze real-world case studies to understand the application of prompt engineering.	K3
CO 5	Design hands-on labs to experiment with prompt engineering using AI tools	K2
DETAILED SYLLABUS		**3-0-0**

Unit	Topic	Proposed Lecture
I	Introduction to Prompt Engineering - Definition and scope of prompt engineering - History and evolution in AI applications - Basic concepts and terminologies - Overview of Large Language Models (LLMs) and their relevance - Applications of prompt engineering in AI - Ethical considerations in prompt engineering	**08**
II	**Designing Effective Prompts for AI Models** Techniques for crafting effective prompts Prompt types: zeroshot, fewshot, and chain of thought Impact of different prompt styles on model output Practical exercises on designing prompts for specific tasks Evaluating prompts: Metrics and techniques Common pitfalls and best practices in prompt design	**08**
III	**Optimization Techniques in Prompt Engineering** Techniques to optimize prompt performance Understanding model biases and mitigating through prompts Case studies on optimization strategies Iterative approaches to prompt refinement Tools and platforms for prompt testing Integrating feedback loops in prompt engineering	**08**
IV	**Real-world Applications and Case Studies** Case studies from various industries Application of prompt engineering in different domains Analysis of successful and failed prompt engineering strategies Role of prompts in interactive AI systems Developing domain-specific prompts Evaluating business impact of prompt engineering	**08**
V	**Analysis of AI Models and Prompt Integration:** Understand how different AI models integrate with prompts theoretically, including the use of GPT and other language models in diverse applications. **Collaboration and Peer Review**: Learn about the role of collaboration and peer review in the theoretical development and refinement of prompt engineering strategies.	**08**

	Evaluating and Presenting Prompt Strategies: understand how to present prompt strategies and incorporate feedback to improve design and application in various AI contexts.	

Text books:

1. Prompt Engineering for Generative AI Book by James Phoenix and Mike Taylor
2. Prompt Engineering for AI Techniques, Strategies, and Best Practice by Dr.Islam Abo Amna
3. Prompt Engineering for Everyone: A Comprehensive Guide to Unlock the Potential of ChatGPT and AI-Language Models by David Scott Bernstein

KAI075	Data Warehousing and Data Mining	
Course Outcome (CO)		**Bloom's Knowledge Level (KL)**
At the end of course , the student will be able to understand		
CO 1	Be familiar with mathematical foundations of data mining tools..	K1 , K2
CO 2	Understand and implement classical models and algorithms in data warehouses and data mining	K3
CO 3	Characterize the kinds of patterns that can be discovered by association rule mining, classification and clustering.	K1 , K2
CO 4	Master data mining techniques in various applications like social, scientific and environmental context.	K3
CO 5	Develop skill in selecting the appropriate data mining algorithm for solving practical problems.	K1 , K2

DETAILED SYLLABUS		**3-0-0**
Unit	**Topic**	**Proposed Lecture**
I	**Data Warehousing:** Overview, Definition, Data Warehousing Components, Building a Data Warehouse, Warehouse Database, Mapping the Data Warehouse to a Multiprocessor Architecture, Difference between Database System and Data Warehouse, Multi Dimensional Data Model, Data Cubes, Stars, Snow Flakes, Fact Constellations, Concept	**08**
II	**Data Warehouse Process and Technology:** Warehousing Strategy, Warehouse /management and Support Processes, Warehouse Planning and Implementation, Hardware and Operating Systems for Data Warehousing, Client/Server Computing Model & Data Warehousing. Parallel Processors & Cluster Systems, Distributed DBMS implementations, Warehousing Software, Warehouse Schema Design,	**08**
III	**Data Mining:** Overview, Motivation, Definition & Functionalities, Data Processing, Form of Data Pre-processing, Data Cleaning: Missing Values, Noisy Data, (Binning, Clustering, Regression, Computer and Human inspection), Inconsistent Data, Data Integration and Transformation. Data Reduction:-Data Cube Aggregation, Dimensionality reduction, Data Compression, Numerosity Reduction, Discretization and Concept hierarchy generation, Decision Tree.	**08**
IV	**Classification**: Definition, Data Generalization, Analytical Characterization, Analysis of attribute relevance, Mining Class comparisons, Statistical measures in large Databases, Statistical-Based Algorithms, Distance-Based Algorithms, Decision Tree-Based Algorithms. Clustering: Introduction, Similarity and Distance Measures, Hierarchical and Partitional Algorithms. Hierarchical Clustering-CURE and Chameleon. Density Based Methods-DBSCAN, OPTICS. Grid Based Methods- STING, CLIQUE. Model Based Method –Statistical Approach, Association rules: Introduction, Large Item sets, Basic Algorithms, Parallel and Distributed Algorithms, Neural Network approach.	**08**
V	**Data Visualization and Overall Perspective:** Aggregation, Historical information, Query Facility, OLAP function and Tools. OLAP Servers, ROLAP, MOLAP, HOLAP, Data Mining interface, Security, Backup and Recovery, Tuning Data Warehouse, Testing Data Warehouse. Warehousing applications and Recent Trends: Types of Warehousing Applications, Web Mining, Spatial Mining and Temporal Mining	**08**

Text books:

1. Alex Berson, Stephen J. Smith "Data Warehousing, Data-Mining & OLAP", TMH
2. Mark Humphries, Michael W. Hawkins, Michelle C. Dy, " Data Warehousing: Architecture and Implementation", Pearson
3. Margaret H. Dunham, S. Sridhar,"Data Mining:Introductory and Advanced Topics" Pearson Education
4. Arun K. Pujari, "Data Mining Techniques" Universities Press
5. Pieter Adriaans, Dolf Zantinge, "Data-Mining", Pearson Education

KAI076	Time series analysis and Forecasting	
Course Outcome (CO)		**Bloom's Knowledge Level (KL)**
At the end of course , the student will be able to understand		
CO 1	Analyze any time series data using various statistical approaches.	K2 , K3
CO 2	Know basic concepts of univariate time series analysis; build appropriate econometric time series models.	K3, K4
CO 3	Know basic concepts of multivariate time series analysis; build appropriate econometric time series models.	K1 , K2
CO 4	Understand limitation and relevance of the models.	K1 , K2
CO 5	Generate reasonable forecast values, and to make concise decisions based on forecasts obtained	K2

DETAILED SYLLABUS		**3-0-0**
Unit	**Topic**	**Proposed Lecture**
I	**INTRODUCTION OF TIMESERIES ANALYSIS**: Introduction to Time Series and Forecasting, Different types of data, Internal structures of time series. Models for time series analysis, Autocorrelation and Partial autocorrelation. Examples of Time series Nature and uses of forecasting, Forecasting Process, Data for forecasting, Resources for forecasting.	**08**
II	**STATISTICS BACKGROUND FOR FORECASTING**: Graphical Displays, Time Series Plots, Plotting Smoothed Data, Numerical Description of Time Series Data, Use of Data Transformations and Adjustments, General Approach to Time Series Modeling and Forecasting, Evaluating and Monitoring Forecasting Model Performance.	**08**
III	**TIME SERIES REGRESSION MODEL**: Introduction Least Squares Estimation in Linear Regression Models, Statistical Inference in Linear Regression, Prediction of New Observations, Model Adequacy Checking, Variable Selection Methods in Regression, Generalized and Weighted Least Squares, Regression Models for General Time Series Data, Exponential Smoothing, First order and Second order.	**08**
IV	**AUTOREGRESSIVE INTEGRATED MOVING AVERAGE (ARIMA) MODELS**: Autoregressive Moving Average (ARMA) Models – Stationary and Inevitability of ARMA Models - Checking for Stationary using Variogram- Detecting Non-stationary - Autoregressive Integrated Moving Average (ARIMA) Models - Forecasting using ARIMA - Seasonal Data -Seasonal ARIMA Models Forecasting using Seasonal ARIMA Models Introduction - Finding the "BEST" Model -Example: Internet Users Data Model Selection Criteria - Impulse Response Function to Study the Differences in Models Comparing Impulse Response Functions for Competing Models .	**08**
V	**MULTIVARIATE TIME SERIES MODELS AND FORECASTING**: Multivariate Time Series Models and Forecasting, Multivariate Stationary Process, Vector ARIMA Models, Vector AR (VAR) Models, Neural Networks and Forecasting Spectral Analysis, Bayesian Methods in Forecasting.	**08**

Text books:

1. Introduction To Time Series Analysis And Forecasting, 2nd Edition, Wiley Series In Probability And Statistics, By Douglas C. Montgomery, Cheryl L. Jen(2015)
2. Master Time Series Data Processing, Visualization, And Modeling Using Python Dr. Avishek Pal Dr. Pks Prakash (2017)
3. Kendall M.G. (1976): Time Series, Charles Griffin.
4. Chatfield C. (1980): The Analysis of Time Series –An Introduction, Chapman & Hall.
5. Mukhopadhyay P. (2011): Applied Statistics, 2nd ed. Revised reprint, Books and Allied

KAI078	Nature-Inspired Computing	
Course Outcome (CO)		**Bloom's Knowledge Level (KL)**
At the end of course , the student will be able :		
CO 1	The basics of Natural systems	K_1 , K_2
CO 2	The concepts of Natural systems and its applications	K_1 , K_2
CO 3	Basic Natural systems functions(operations)	K_2
CO 4	Natural design considerations.	K_2 , K_3
CO 5	Integration of Hardware and software in Natural applications.	K_3, K_6

DETAILED SYLLABUS		**3-0-0**
Unit	**Topic**	**Proposed Lecture**
I	INTRODUCTION: From Nature to Nature Computing , Philosophy , Three Branches: A Brief Overview, Individuals, Entities and agents - Parallelism and Distributivity Interactivity ,Adaptation Feedback-Self-Organization-Complexity, Emergence and ,Bottom-up Vs Top-Down-Determination, Chaos and Fractals	**08**
II	Computing Inspired by Nature: Evolutionary Computing, Hill Climbing and Simulated Annealing, Darwin's Dangerous Idea, Genetics Principles, Standard Evolutionary Algorithm –Genetic Algorithms , Reproduction-Crossover, Mutation, Evolutionary Programming, Genetic Programming	**08**
III	SWARM INTELLIGENCE: Introduction - Ant Colonies, Ant Foraging Behavior, Ant Colony Optimization, SACO and scope of ACO algorithms, Ant Colony Algorithm (ACA), Swarm Robotics, Foraging for food, Social Adaptation of Knowledge , Particle Swarm Optimization (PSO)	**08**
IV	IMMUNOCOMPUTING: Introduction- Immune System, Physiology and main components, Pattern Recognition and Binding , Immune Network Theory- Danger Theory, Evaluation Interaction-Immune Algorithms , Introduction – Genetic algorithms , Bone Marrow Models ,Forest's Algorithm, Artificial Immune Networks	**08**
V	COMPUTING WITH NEW NATURAL MATERIALS: DNA Computing: Motivation, DNA Molecule , Adleman's experiment , Test tube programming language, Universal DNA Computers , PAM Model , Splicing Systems , Lipton's Solution to SAT Problem , Scope of DNA Computing , From Classical o DNA Computing	**08**

Text books:

1. Leandro Nunes de Castro, " Fundamentals of Natural Computing, Basic Concepts, Algorithms and

2. Applications", Chapman & Hall/ CRC, Taylor and Francis Group, 2007

3. Floreano D. and Mattiussi C., "Bio-Inspired Artificial Intelligence: Theories, Methods, and Technologies",

MIT Press, Cambridge, MA, 2008.

4. Albert Y.Zomaya, "Handbook of Nature-Inspired and Innovative Computing", Springer, 2006.

5. Marco Dorrigo, Thomas Stutzle," Ant Colony Optimization", PHI,2005

KDS078	Deep Learning	
Course Outcome (CO)		**Bloom's Knowledge Level (KL)**
At the end of course , the student will be able :		
CO 1	To present the mathematical, statistical and computational challenges of building neural networks	K_1 , K_2
CO 2	To study the concepts of deep learning	K_1 , K_2
CO 3	To introduce dimensionality reduction techniques	K_2
CO 4	To enable the students to know deep learning techniques to support real-time applications	K_2 , K_3
CO 5	To examine the case studies of deep learning techniques	K_3, K_6

DETAILED SYLLABUS		**3-0-0**
Unit	**Topic**	**Proposed Lecture**
I	**INTRODUCTION** : Introduction to machine learning- Linear models (SVMs and Perceptrons, logistic regression)- Intro to Neural Nets: What a shallow network computes- Training a network: loss functions, back propagation and stochastic gradient descent- Neural networks as universal function approximates	**08**
II	**DEEP NETWORKS** : History of Deep Learning- A Probabilistic Theory of Deep Learning- Backpropagation and regularization, batch normalization- VC Dimension and Neural Nets-Deep Vs Shallow Networks-Convolutional Networks- Generative Adversarial Networks (GAN), Semi-supervised Learning	**08**
III	**DIMENTIONALITY REDUCTION** 9 Linear (PCA, LDA) and manifolds, metric learning - Auto encoders and dimensionality reduction in networks - Introduction to Convnet - Architectures – AlexNet, VGG, Inception, ResNet - Training a Convnet: weights initialization, batch normalization, hyperparameter optimization	**08**
IV	**OPTIMIZATION AND GENERALIZATION** : Optimization in deep learning– Non-convex optimization for deep networks- Stochastic Optimization Generalization in neural networks- Spatial Transformer Networks- Recurrent networks, LSTM - Recurrent Neural Network Language Models- Word-Level RNNs & Deep Reinforcement Learning - Computational & Artificial Neuroscience	**08**
V	**CASE STUDY AND APPLICATIONS** : Imagenet- Detection-Audio WaveNet-Natural Language Processing Word2Vec - Joint Detection-Bioinformatics- Face Recognition- Scene Understanding- Gathering Image Captions	**08**

Text books:

1. Cosma Rohilla Shalizi, Advanced Data Analysis from an Elementary Point of View, 2015.

2. Deng & Yu, Deep Learning: Methods and Applications, Now Publishers, 2013.

3. Ian Goodfellow, Yoshua Bengio, Aaron Courville, Deep Learning, MIT Press, 2016.

4. Michael Nielsen, Neural Networks and Deep Learning, Determination Press, 2015.

Mapping with MOOCS:

https://onlinecourses.nptel.ac.in/noc18_cs41/preview

KDS079	Service Oriented Architecture	
	Course Outcome (CO)	**Bloom's Knowledge Level (KL)**
	At the end of course , the student will be able :	
CO 1	Comprehend the need for SOA and its systematic evolution.	K1 , K2
CO 2	Apply SOA technologies to enterprise domain.	K3
CO 3	Design and analyze various SOA patterns and techniques.	K4
CO 4	Compare and evaluate best strategies and practices of SOA.	K2
CO 5	Understand the business case for SOA	K1

DETAILED SYLLABUS		**3-0-0**
Unit	**Topic**	**Proposed Lecture**
I	**Introduction:** SOA and MSA Basics: Service Orientation in Daily Life, Evolution of SOA and MSA. Serviceoriented Architecture and Microservices architecture – Drivers for SOA, Dimensions of SOA, Conceptual Model of SOA, Standards and Guidelines for SOA, Emergence of MSA. **Enterprise-Wide SOA:** Considerations for Enterprise-wide SOA, Strawman Architecture for Enterprise-wide SOA, Enterprise SOA Reference Architecture, Object-oriented Analysis and Design (OOAD) Process, Service-oriented Analysis and Design (SOAD) Process, SOA Methodology for Enterprise	**08**
II	**Service-Oriented Applications:** Considerations for Service-oriented Applications, Patterns for SOA, Pattern-based Architecture for Service-oriented Applications, Composite Applications, Composite Application Programming Model. **Service-Oriented Analysis and Design:** Need for Models, Principles of Service Design, Nonfunctional Properties for Services, Design of Activity Services (or Business Services), Design of Data Services, Design of Client Services, Design of Business Process Services.	**08**
III	**Technologies for SOA:** Technologies for Service Enablement, Technologies for Service Integration, Technologies for Service Orchestration. **SOA Governance and Implementation:** Strategic Architecture Governance, Service Design-time Governance, Service Run-time Governance, Approach for Enterprise-wide SOA Implementation.	**08**
IV	**Big Data and SOA:** Concepts, Big Data and its characteristics, Technologies for Big Data, Service-orientation for Big Data Solutions. **Business Case for SOA:** Stakeholder Objectives, Benefits of SOA, Cost Savings, Return on Investment (ROI), Build a Case for SOA	**08**
V	**SOA Best Practices:** SOA Strategy – Best Practices, SOA Development – Best Practices, SOA Governance – Best Practices. **EA and SOA for Business and IT Alignment:** Enterprise Architecture, Need for Business and It Alignment, EA and SOA for Business and It Alignment	**08**

Text books:

Shankar Kambhampaty; Service - Oriented Architecture & Microservices Architecture: For Enterprise, Cloud, Big Data and Mobile; Wiley; 3rd Edition; 2018; ISBN: 9788126564064.

Icon Group International; The 2018-2023 World Outlook for Service-Oriented Architecture (SOA) Software and Services;

ICON Group International; 1st Edition, 2017; ASIN: B06WGPN8YD.

Thomas Erl; Service Oriented Architecture Concepts Technology & Design; Pearson Education Limited; 2015; ISBN-13: 9788131714904.

Guido Schmutz, Peter Welkenbach, Daniel Liebhart; Service Oriented Architecture An Integration Blueprint; Shroff Publishers & Distributors; 2010; ISBN-13: 9789350231081

KAD072	Large Language Models (LLMs) and Their Applications	
Course Outcome (CO)		**Bloom's Knowledge Level (KL)**
At the end of course , the student will be able to understand		
CO 1	Understand the architecture and foundational concepts of LLMs.	K_1 , K_2
CO 2	Explore various LLMs like GPT, BERT, and T5.	K_2 , K_3
CO 3	Apply LLMs in NLP tasks, chatbots, and content generation.	K_2 , K_3
CO 4	Evaluate the performance and limitations of different LLMs.	K_3 , K_4
CO 5	Implement practical applications using LLMs.	K_3, K_6
DETAILED SYLLABUS		**3-0-0**

Unit	Topic	Proposed Lecture
I	Introduction to Large Language Models Overview of Large Language Models (LLMs) Historical development and evolution of LLMs Basic concepts and terminologies in LLMs Architectural overview of models like GPT, BERT, and T5 Applications of LLMs in various domains Ethical considerations in the use of LLMs	**08**
II	Architectures of GPT, BERT, T5, and Others Detailed architecture of GPT: Generative Pretrained Transformer Understanding BERT: Bidirectional Encoder Representations from Transformers Exploring T5: Text-to-Text Transfer Transformer Comparative analysis of different LLM architectures Training and fine-tuning of LLMs Limitations and challenges in current LLM architectures	**08**
III	Applications in NLP and Chatbots Utilizing LLMs for Natural Language Understanding (NLU) Building conversational agents and chatbots using LLMs Text generation, summarization, and translation using LLMs Case studies: Successful deployments of LLM-based systems Hands-on projects: Developing NLP applications Challenges in deploying LLMs in production environments	**08**
IV	Case Studies and Performance Evaluation Case studies on the use of LLMs in various industries Performance evaluation metrics for LLMs Understanding biases and fairness in LLM outputs Techniques for improving LLM performance Evaluating the impact of LLMs on society and industry Future trends and developments in LLMs	**08**

V	**Introduction to LLM Architecture:**. • **Supported by Hugging Face** GPT, BERT, T5, and other models available in the Hugging Face Model Hub. • **Customizing LLM Architectures:** Techniques for modifying model architectures for specific use cases. • **Transfer Learning with LLMs:** Leveraging transfer learning for domain adaptation and enhancing model performance. • **Integration with Other Frameworks:** Using Hugging Face in conjunction with other frameworks like PyTorch and TensorFlow.	**08**

Text books:

1. Mastering Large Language Models: Advanced techniques, applications, cutting-edge methods, and top LLMs by Sanket Subhash Khandare (Author)
2. Mastering Large Language Models with Python: Unleash the Power of Advanced Natural Language Processing for Enterprise Innovation and Efficiency Using ... Models (LLMs) with Python by Raj Arun R (Author)
3. Transformers for Natural Language Processing by Denis Rothman
4. Natural Language Processing with PyTorch by Delip Rao and Brian McMahan

KAD073	Robotics and Automation	
	Course Outcome (CO)	**Bloom's Knowledge Level (KL)**
	At the end of course, the student will be able to understand	
CO 1	Understand the integration of machine learning techniques with network security.	K1
CO 2	Analyze different network threats and apply machine learning models for threat detection.	K2
CO 3	Implement supervised and unsupervised machine learning algorithms for anomaly detection in network traffic.	K3
CO 4	Develop systems for predicting and mitigating security breaches using advanced machine learning techniques.	K3
CO 5	Evaluate the effectiveness and scalability of machine learning models in diverse network environments.	K4

DETAILED SYLLABUS		**3 1 0**
Unit	**Topic**	**Proposed Lecture**
I	**Introduction to Machine Learning in Network Security:** Overview of machine learning techniques, Introduction to network security, Integration of machine learning in network security.	**08**
II	**Anomaly Detection and Intrusion Detection Systems (IDS):** Machine Learning Models for Anomaly Detection, Supervised and Unsupervised Learning Techniques, Data Preprocessing, Types of IDS, Implementing IDS using Machine Learning, Case Studies of Machine Learning based IDS, Challenges and Future Directions.	**08**
III	**Malware Analysis and Network Traffic Analysis:** Types of Malware, Feature Extraction Techniques, Machine Learning Models for Malware Detection and Classification, Hands-on with Malware Datasets, Machine Learning for Network Traffic Classification, Predictive Modeling, Real-time Analysis Techniques, Building a Predictive Model for Network Attack Classification.	**08**
IV	**Securing the Consumer Web:** Monetizing the Consumer Web, Data Abuse and Preventive Measures, Machine Learning for Abuse Detection, Case Studies and Applications.	**08**
V	**Production Systems in Machine Learning for Network Security:** Maturity and Scalability of ML Systems, Data Quality and Model Quality, Performance Metrics, Security and Reliability in Machine Learning Systems.	**08**

Text books:

1. Clarence Chio, David Freeman, *Machine Learning and Security*, O'Reilly Media.
2. Sumeet Dua, Xian Du, *Data Mining and Machine Learning in Cybersecurity*, CRC Press.
3. Himanshu Kumar, *Learning Resources for Penetration Testing.*
4. *The Web Application Hacker's Handbook: Finding and Exploiting Security Flaws*, 2nd Edition.
5. Prakhar Prasad, *Mastering Modern Web Penetration Testing.*

eBooks:

6. Charlie Kaufman, Radia Perlman, Mike Speciner, *Network Security: Private Communication in a Public World.*
7. Chris Sanders, *Practical Packet Analysis: Using Wireshark to Solve Real-World Network Problems.*
8. William Stallings, *Network Security Essentials: Applications and Standards.*

MOOC/ Video Lectures Available At:

9. *Machine Learning* by Stanford University on Coursera.
10. *Deep Learning Specialization* by deeplearning.ai on Coursera.
11. *Machine Learning with Python* by IBM on Coursera.

KAD074	Cognitive Computing	
Course Outcome (CO)		**Bloom's Knowledge Level (KL)**
At the end of course, the student will be able to understand		
CO 1	Understand the foundation and principles of cognitive computing and its applications.	K1,K2
CO 2	Apply design principles to build cognitive systems and leverage machine learning for hypothesis generation and scoring.	K2
CO 3	Utilize natural language processing (NLP) techniques to support cognitive systems and solve business problems.	K4
CO 4	Effectively represent knowledge using taxonomies and ontologies in cognitive systems.	K3
CP 5	Explore the relationship between big data and cognitive computing, and integrate big data with traditional data sources.	K_2, K_3

DETAILED SYLLABUS		**3 1 0**
Unit	**Topic**	**Proposed Lecture**
I	**Foundation of Cognitive Computing:** Cognitive computing as a new generation, the uses of cognitive systems, system cognitive, gaining insights from data, Artificial Intelligence as the foundation of cognitive computing, understanding cognition.	**08**
II	**Design Principles for Cognitive Systems:** Components of a cognitive system, building the corpus, bringing data into cognitive systems, machine learning, hypotheses generation and scoring, presentation and visualization services.	**08**
III	**Natural Language Processing (NLP) in Cognitive Systems:** Role of NLP in a cognitive system, semantic web, Applying NLP technologies to business problems, representing knowledge.	**08**
IV	**Knowledge Representation in Cognitive Systems:** Defining taxonomies and ontologies, knowledge representation models, implementation considerations, effective representation of knowledge in cognitive systems.	**08**
V	**Big Data and Advanced Analytics in Cognitive Computing:** Relationship between big data and cognitive computing, integrating big data with traditional data sources, using advanced analytics techniques, the impact of open source tools on cognitive computing.	**08**

Text books:

1. Judith H. Hurwitz, Marcia Kaufman, Adrian Bowles, *Cognitive Computing and Big Data Analytics*, Wiley.

eBooks:

2. Peter Fingar, *Cognitive Computing: A Brief Guide for Game Changers*. This book provides an overview of cognitive computing and its potential applications in various industries.

MOOC/ Video Lectures Available At:

3. *Introduction to Cognitive Computing* by IBM on Coursera. This course offers an introduction to cognitive computing, covering topics such as natural language processing, machine learning, and advanced analytics.

KCS710	Quantum Computing	
	Course Outcome (CO)	**Bloom's Knowledge Level (KL)**
	At the end of course , the student will be able to understand	
CO 1	Distinguish problems of different computational complexity and explain why certain problems are rendered tractable by quantum computation with reference to the relevant concepts in quantum theory.	K_1 , K_2
CO 2	Demonstrate an understanding of a quantum computing algorithm by simulating it on a classical computer, and state some of the practical challenges in building a quantum computer.	K_2 , K_3
CO 3	Contribute to a medium-scale application program as part of a co-operative team, making use of appropriate collaborative development tools (such as version control systems).	K_2 , K_3
CO 4	Produce code and documentation that is comprehensible to a group of different programmers and present the theoretical background and results of a project in written and verbal form.	K_3 , K_4
CO 5	Apply knowledge, skills, and understanding in executing a defined project of research, development, or investigation and in identifying and implementing relevant outcomes.	K_3, K_6
	DETAILED SYLLABUS	**3-0-0**

Unit	Topic	Proposed Lecture
I	**Fundamental Concepts:** Global Perspectives, Quantum Bits, Quantum Computation, Quantum Algorithms, Quantum Information, Postulates of Quantum Mechanisms.	08
II	**Quantum Computation**: Quantum Circuits – Quantum algorithms, Single Orbit operations, Control Operations, Measurement, Universal Quantum Gates, Simulation of Quantum Systems, Quantum Fourier transform, Phase estimation, Applications, Quantum search algorithms – Quantum counting – Speeding up the solution of NP – complete problems – Quantum Search for an unstructured database.	08
III	**Quantum Computers:** Guiding Principles, Conditions for Quantum Computation, Harmonic Oscillator Quantum Computer, Optical Photon Quantum Computer – Optical cavity Quantum electrodynamics, Ion traps, Nuclear Magnetic resonance	08
IV	**Quantum Information:** Quantum noise and Quantum Operations – Classical Noise and Markov Processes, Quantum Operations, Examples of Quantum noise and Quantum Operations – Applications of Quantum operations, Limitations of the Quantum operations formalism, Distance Measures for Quantum information.	08
V	**Quantum Error Correction:** Introduction, Shor code, Theory of Quantum Error –Correction, Constructing Quantum Codes, Stabilizer codes, Fault – Tolerant Quantum Computation, Entropy and information – Shannon Entropy, Basic properties of Entropy, Von Neumann, Strong Sub Additivity, Data Compression, Entanglement as a physical resource .	08

Text books:

1. Micheal A. Nielsen. &Issac L. Chiang, "Quantum Computation and Quantum Information", Cambridge University Press, Fint South Asian edition, 2002.

2. Eleanor G. Rieffel , Wolfgang H. Polak , "Quantum Computing - A Gentle Introduction" (Scientific and Engineering Computation) Paperback – Import,

3 Oct 2014 3. Computing since Democritus by Scott Aaronson

4. Computer Science: An Introduction by N. DavidMermin 5. Yanofsky's and Mannucci, Quantum Computing for Computer Scientists.

KAD075	Machine Learning & Network Security	
	Course Outcome (CO)	**Bloom's Knowledge Level (KL)**
	At the end of course, the student will be able to	
CO 1	Learn different machine learning algorithms to secure information.	K_1, K_2
CO 2	Implement filtering methods using machine learning techniques.	K_2, K_3
CO 3	Analyze different methods of detecting anomalies.	K_3, K_4
CO 4	Perform malware analysis using extracted information.	K_4, K_5
CO 5	Visualize the attacks on consumer websites.	K_3, K_4
CO 6	Model machine learning-based systems to create production environments.	K_3, K_4

DETAILED SYLLABUS		**3 0 0**
Unit	**Topic**	**Proposed Lecture**
I	**Convergence of Machine Learning and Network Security:** Cyber Threat Landscape, The Cyber Attacker's Economy, Overview of Machine Learning, Real World Uses of Machine Learning in Security, Spam Fighting: An Iterative Approach.	**08**
II	**Anomaly Detection and Network Traffic Analysis:** Anomaly Detection vs. Supervised Learning, Intrusion Detection with Heuristics, Data Driven Methods, Feature Engineering for Anomaly Detection, Challenges of Using Machine Learning in Anomaly Detection, Theory of Network Defense, Building a Predictive Model to Classify Network Attacks.	**08**
III	**Malware Analysis and Protecting the Consumer Web:** Understanding Malware, Feature Generation, From Features to Classification, Live and Dead Malware Analysis, Android Malware Analysis, Monetizing the Consumer Web, Types of Abuse and the Data That Can Stop Them, Supervised Learning for Abuse Problems, Clustering Abuse.	**08**
IV	**Machine Learning Algorithms for Security Applications:** Implementing Filtering Methods, Different Methods of Detecting Anomalies, Visualization of Attacks on Consumer Websites, Practical Applications in Speech Recognition, Image Recognition, and Target Recognition.	**08**
V	**Production Systems and Advanced Machine Learning Techniques:** Defining Machine Learning System Maturity and Scalability, Data Quality, Model Quality, Performance, Maintainability, Monitoring and Alerting, Security and Reliability, Translating Machine Learning Algorithms from Lab to Production.	**08**

Text books:

1. Clarence Chio, David Freeman, *Machine Learning and Security*, O'Reilly Media, Inc. ISBN: 9781491979907.
2. Sumeet Dua, Xian Du, *Data Mining and Machine Learning in Cybersecurity*, CRC Press, ISBN: 9781439839423.
3. Himanshu Kumar, *Learning Nessus for Penetration Testing*.
4. *The Web Application Hacker's Handbook: Finding and Exploiting Security Flaws*, 2nd Edition.
5. Prakhar Prasad, *Mastering Modern Web Penetration Testing*.

eBooks:

6. Charlie Kaufman, Radia Perlman, Mike Speciner, *Network Security: Private Communication in a Public World*.
7. Chris Sanders, *Practical Packet Analysis: Using Wireshark to Solve Real World Network Problems*.
8. William Stallings, *Network Security Essentials: Applications and Standards.*

MOOC/ Video Lectures Available At:

9. *Machine Learning* by Stanford University on Coursera.
10. *Deep Learning Specialization* by deeplearning.ai on Coursera.
11. *Machine Learning with Python* by IBM on Coursera.

KCS354/KCS554/KCS752	Mini Project or Internship Assessment	
Course Outcome (CO)		**Bloom's Knowledge Level (KL)**
At the end of course , the student will be able to understand		
CO 1	Developing a technical artifact requiring new technical skills and effectively utilizing a new software tool to complete a task	K_4 , K_5
CO 2	Writing requirements documentation, Selecting appropriate technologies, identifying and creating appropriate test cases for systems.	K_5 , K_6
CO 3	Demonstrating understanding of professional customs & practices and working with professional standards.	K_4 , K_5
CO 4	Improving problem solving, critical thinking skills and report writing.	K_4 , K_5
CO 5	Learning professional skills like exercising leadership, behaving professionally, behaving ethically, listening effectively, participating as a member of a team, developing appropriate workplace attitudes.	K_2, K_4

KCS 753/ KCS851	Project	
Course Outcome (CO)		**Bloom's Knowledge Level (KL)**
At the end of course , the student will be able to understand		
CO 1	Analyze and understand the real life problem and apply their knowledge to get programming solution.	K_4 , K_5
CO 2	Engage in the creative design process through the integration and application of diverse technical knowledge and expertise to meet customer needs and address social issues.	K_4 , K_5
CO 3	Use the various tools and techniques, coding practices for developing real life solution to the problem.	K_5 , K_6
CO 4	Find out the errors in software solutions and establishing the process to design maintainable software applications	K_4 , K_5
CO 5	Write the report about what they are doing in project and learning the team working skills	K_5, K_6